How to Print a set of Multi-Material Bellows:

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1. Create Geometry in SolidWorks

a. Download existing SolidWorks Files

The most up to date CAD models are located in the MACLAB one drive under MacCurdy Lab/Lab Projects/Printable Hydraulics Designs/SolidWorks Files/. Download the folder 10.13_Variable_Bellows_VeroLayer for a single bellows stack.

b. Generate .STL files

To open the bellows models in OpenVCad you will need to create a .STL file of each individual model in the full bellows assembly. Pre-generated .STL files are in a .zip file called “variable_bellows_STL.zip”.

If you want to generate the .STL files yourself, start by opening the top-level bellows assembly (variableNumberBellowsStack_ASSEMBLY.SLDASM). You should see the model shown below:

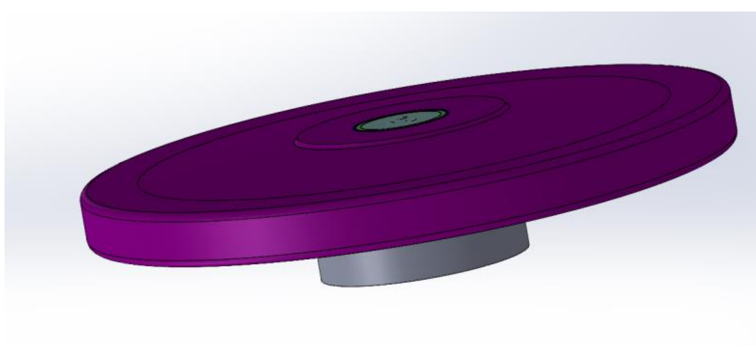
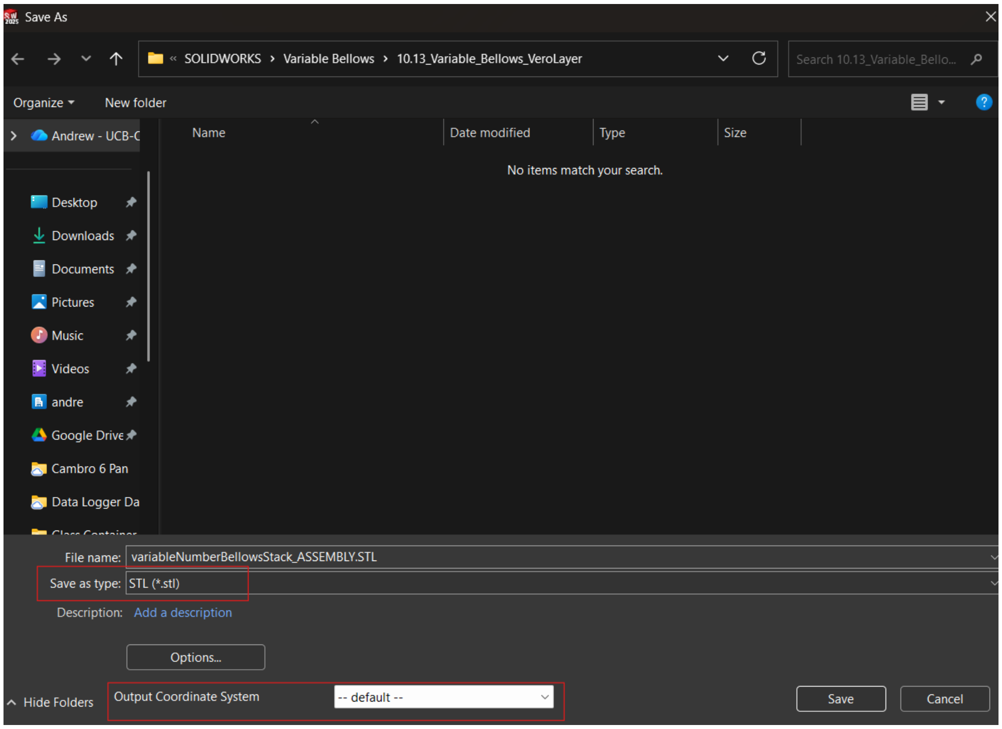


Fig. 1: variableNumberBellowsStack_ASSEMBLY.SLDASM

This assembly has individual part models for the outer bellows “shell”, the top & bottom bellows caps, a keep-out layer of hard *vero* material, a keep-out layer of support material, and the working fluid.

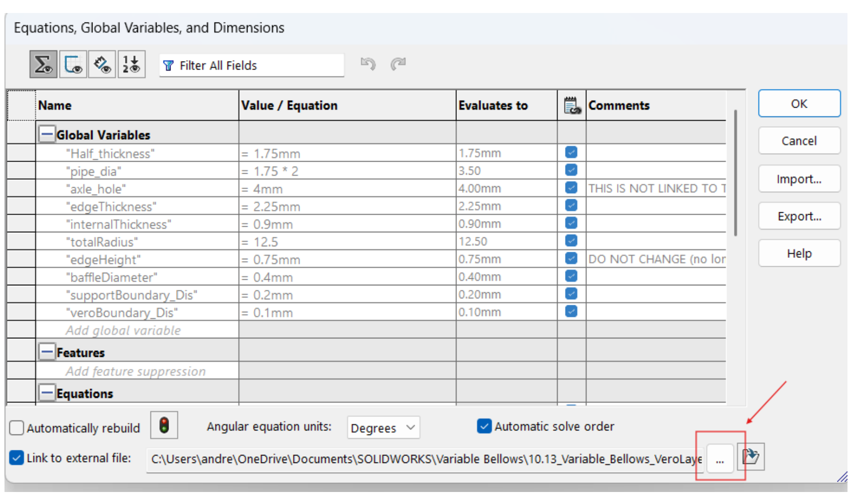
To save the model as .STL files go to options: save as, and then set “save as type:” to .stl, make sure that “Output Coordinate System” is set to default, and then select your desired directory. Make sure that **no parts are suppressed** in the assembly when saving .STL files.



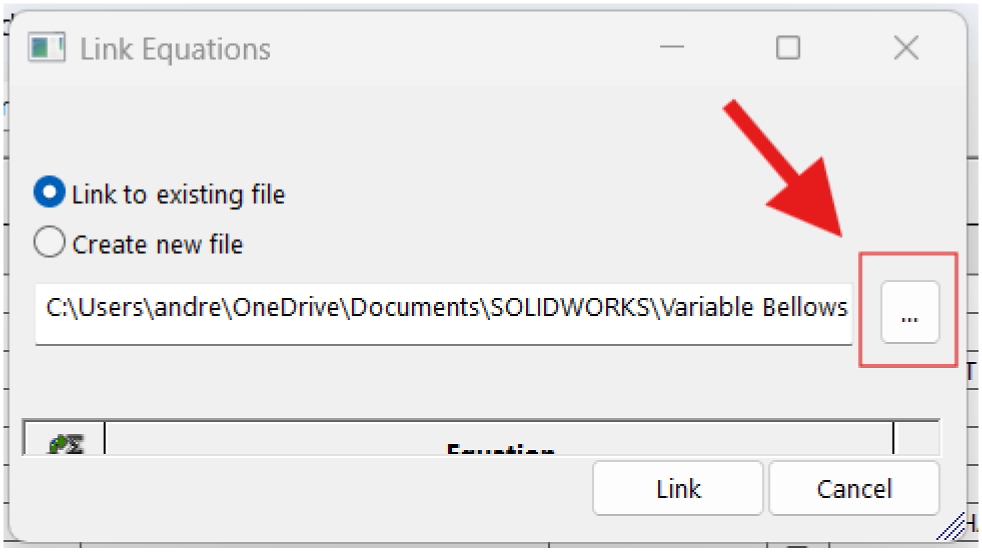
Optional:

If you want to change the geometry of the bellows you can do so by changing values in the file "equations.txt". Each individual part of the assembly is linked to this file. **DO NOT** change the values of the equations in individual SolidWorks parts.

If this is the first time you have downloaded these SolidWorks files you will need to re-link "equations.txt" with each part. Start by opening "Variable_Bellow.SLDPRT" and going to Tools: Equations in the upper left-hand corner. You will now see the following window and will need to select "Edit Link" (the three dots in the bottom right corner).



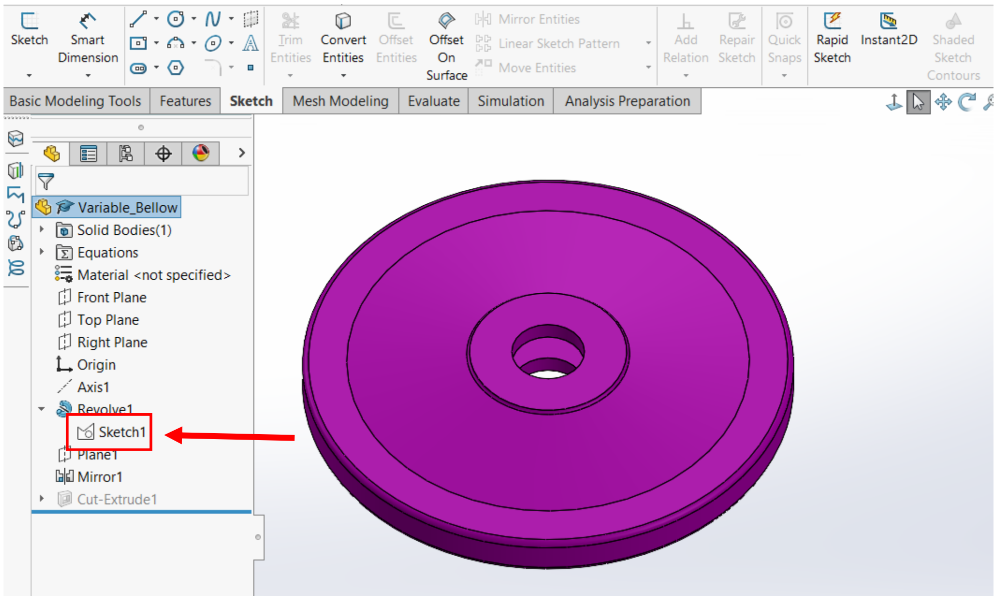
This will open a second window. Select the three dots again and navigate to the directory you downloaded “10.13_Variable_Bellows_VeroLayer” to and select “equations.txt”. Then select Link and hit OK when prompted. You will need to repeat this with “variable_fluid”, “Variable_SupportBundary”, and “Variable_VeroBundary”.



The current geometries are shown below:

```
"Half_thickness"= 1.75mm
"D6@Sketch1"= "Half_thickness"
"pipe_dia"= 1.75 * 2
"axle_hole"= 4mm 'THIS IS NOT LINKED TO THE PART FILE
"edgeThickness"= 2.25mm
"D2@Sketch1"="edgeThickness"
"internalThickness"= 0.9mm
"D7@Sketch1"="internalThickness"
"totalRadius"= 12.5
"D4@Sketch1"="totalRadius"
"edgeHeight"= 0.75mm   'DO NOT CHANGE (no longer relevant, dependent on internalThickness)
"baffleDiameter"= 0.4mm
"supportBoundary_Dis"= 0.2mm
```

When you make a change to a parameter in “equations.txt”, save the .txt file and then in SolidWorks, hit rebuild to update the part or assembly to the new dimension. If you want to know what an individual dimension corresponds to you can open the first sketch on either “variable_bellow”, “variable_fluid”, “Variable_SupportBundary”, or “Variable_VeroBundary”. This sketch is the same for each part, but different sections are selected in “Revolve1” to generate each individual part.



2. Generate OpenVCad Script

a. Download Code from MACLab GitHub.

All code for the bellows design project is located here: https://github.com/MacCurdyLab/multiMaterial_Bellows. To generate the OpenVCad script you will need to download the contents of Fourier_Series_Generator.

Open the file “FouierCalculation2.2_FileGenerator.ipynb” using your text editor of choice. (I use VSCode). This script will generate a gradient of hard and soft plastic for the outer shell of the bellows. It does this by approximating a piecewise gradient function using a Fourier transform because OpenVCad can only use differentiable-continuous functions to create material gradients.

b. Setup “FouierCalculation2.2_FileGenerator.ipynb”

Set *generateFile* to true and set *base_output_folder* to your desired directory (its easiest if this is the same directory that you previously saved the .STL files to):

```
# === File Saving ===
generateFile = True
base_output_folder = r"C:\Git_Repoes\multiMaterial_Bellows\OpenVCad_Scripts\SingleStack"
```

Now you can change the material gradient settings. The following values are the most up to date parameters. If this is your first time generating bellows, it's easiest to leave these values as is.

N dictates how accurate the Fourier approximation of our gradient function is; we've had good results between 60 and 85. If your models are taking a long time to render in OpenVCAD it is most likely that you have too many Fourier nodes.

```
# === GRADIENT SETTINGS ===
use_radial = False   # True for radial plots, False for regular plotss
N = 60               # number of Fourier modes (solid quality at 85)
agilus_stiff_section = 30  # percentage of agilus in the stiff portion of the bellows
vero_stiff_section = 100 - agilus_stiff_section
agilus_soft_section = 40   # percentage of agilus in the soft portion of the bellows  #Last Value: 80
vero_soft_section = 100 - agilus_soft_section
```

We've pre-coded four different gradients. If you're interested in what each gradient looks like you can set *generateFile* to false, select the gradient of your choosing, and then run the script. At the bottom of the script a graph of the gradient function will be generated.

```
# === Choose one gradient type ===
gradient1 = False   # Gradient for constant soft sections
gradient2 = False  # Gradient for linear soft sections
gradient3 = True  # Gradient for parabolic soft sections
gradient4 = False  # Gradient for some vero in the middle cap of the bellows
```

This script will also generate a VCAD script for a dual bellows stack, for this tutorial make sure that *stack* is set to True.

```
# === Design Settings ===
stack = True #single stack of bellows
dual_wb = False #double stack with baffles for test stand
dual_nb = False #double stack without baffles for test stand
```

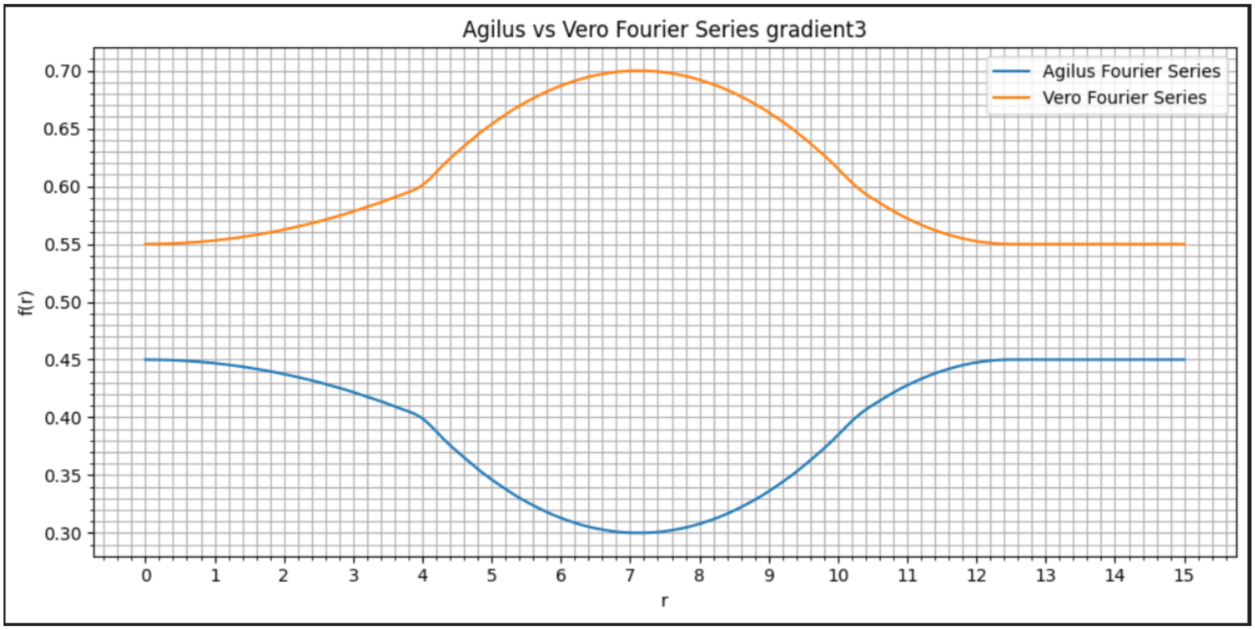
The following Geometry settings need to be the same as the geometry settings you set in *Section 1a* in "equations.txt". The bellow table indicates what values correspond to each other.

Note: This manual updating of geometry values is the primary motivation for switching from designing the CAD models of the bellows in SolidWorks to CadQuery. A preliminary version of this new workflow can be found here.

"equations.txt"	"FouierCalculation2.2_FileGenerator.ipynb"
2 * "totalRadius"	*mainDiameter*
"pipe_dia"	*pipeDiameter*
"edgeThickness"	*edgeThickness*

```
# === Geometry Settings ===
mainDiameter = 25  # diameter of each bellows
pipeDiameter = 3.5  # inner diameter of bellow opening
edgeThickness = 2.25  # thickness of inner and outter sections of each bellows
```

You can now run the script. The gradient function for the bellows shell should look similar to this:



You should get the following confirmation if your VCAD script is successfully saved.

```
Wrote: C:\Git_Repoes\multiMaterial_Bellows\OpenVCad_Scripts\SingleStack\fourier_SingleStack_gradient3_N60_stiff30-70_soft40-60_grad45-40_.py
```

3. Generate Model to Print on J750 3D Printer

a. Download OpenVCAD studio

The easiest way to get started with OpenVCAD is with OpenVCAD studio. This is a text editor that will render your models as you run your scripts.

b. Set up your Bellows Script in OpenVCAD Studio

Open your script directly in OpenVCAD Studio.

Set the file location that you downloaded your .STL files to in *Section 1b* and note the material definitions.

```
#-- STL File Directory --
STL_Location = "path/to/directory"

#-- Material definitions --
materials = pv.default_materials()
red = materials.id("red")             # Agilus
blue = materials.id("blue")           # Vero
liquid_mat = materials.id("liquid")   # Liquid matieral
green = materials.id("green")         # Support
yellow = materials.id("yellow")       # Air
```

Again you need to make sure that the following geometry variables are the same as you set in *Section 1a*. **Do Not** change *BottomcapHeight* or *topCapHeight*.

“equations.txt”	OpenVCAD Script
2 * “totalRadius”	*mainD*
2 * “Half_Thickness”	*mainHeight*

```
#-- Dimensions of part --
mainHeight = 3.5    # Dr. Mac's: 3.5[mm]
mainD = 25          # Dr. Mac's: 25[mm]
BottomcapHeight = 1       #[mm]
topCapHeight = 1.2 #[mm]
```

fluidPercent is the percent fluid in an air-fluid metamaterial. We don’t print 100% fluid since the J750 over jets cleaning fluid when its Head Voltages are calibrated for solid materials.

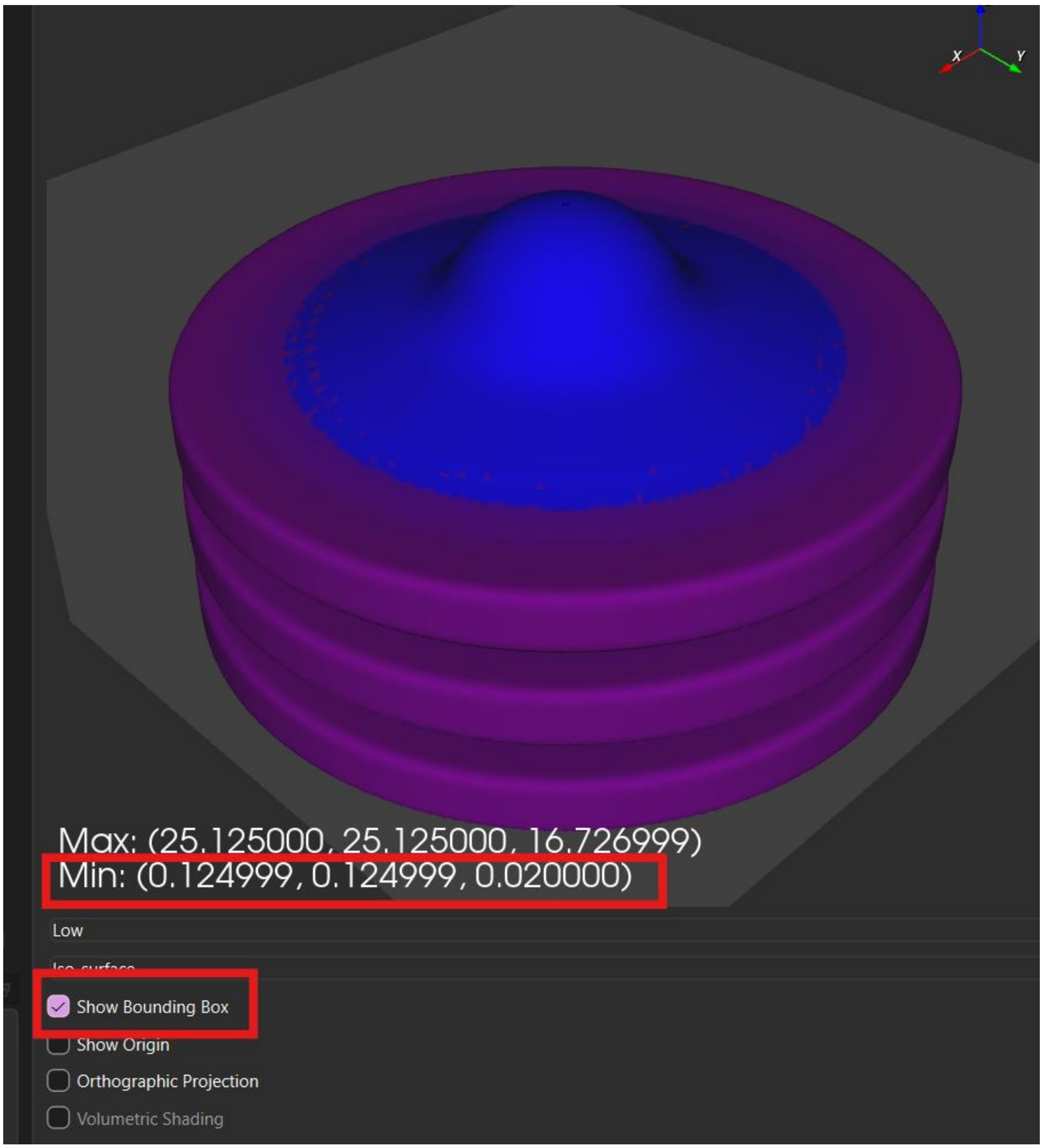
So far we have only printed with *includeBaffles* as true. In a 2016 Paper Dr. MacCurdy found it beneficial to have small baffles (cylidrical support pillars) made of support material in regions of fluid to help support capping layers of solid materials.

NumStacks is how many stacks of bellows high a single stack is, and *num_bellowsToPrint* is how many individual sets of stacks of bellows you want to print.

```
#-- Stack Settings --
fluidPercent = 0.725
numStacks = 3
includeBaffles = True
num_bellowsToPrint = 1
```

You can now render your bellows by going to File:Render. Unfortunately, your material gradient will not be correctly updated until you update the variables *x, y, and z* to be equal to the minimum bounding box values of your model.

```
#-- placing stuff --
x = 0.124999
y = 0.124999
z = 0.02
```



c. Exporting Bellows as a .png Stack:

The J750 takes a folder of .png files with each file representing an individual layer of the print. To generate these .png files go to Export: PNG Stack and select a folder to store the .PNGs in. **DO NOT** edit the Voxel Size settings as these are already set for the resolution of the J750.